TEAM 1
IT ARCHITECTS

CREDIT CARD FRAUD DETECTION

Agenda

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- Improvement from Professor
- Problem Statement
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- Team Working Agreement
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- Product Design
- Technology and Tools
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- Product Backlog
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- Historical and Average Velocities
- Committed and Completed Ratios
- Sprint Retrospective
- Application Screen Shot
- Rest API
- Wiki Page & GitHub Repository
- Project Demo

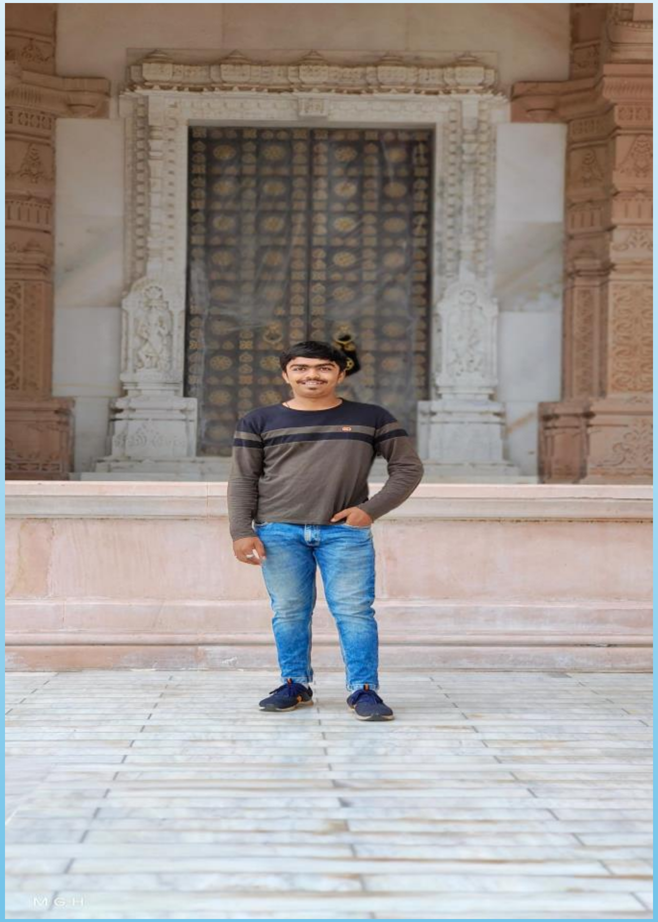
Team Members

Mr. Arth Patel

Role : Front-End Developer

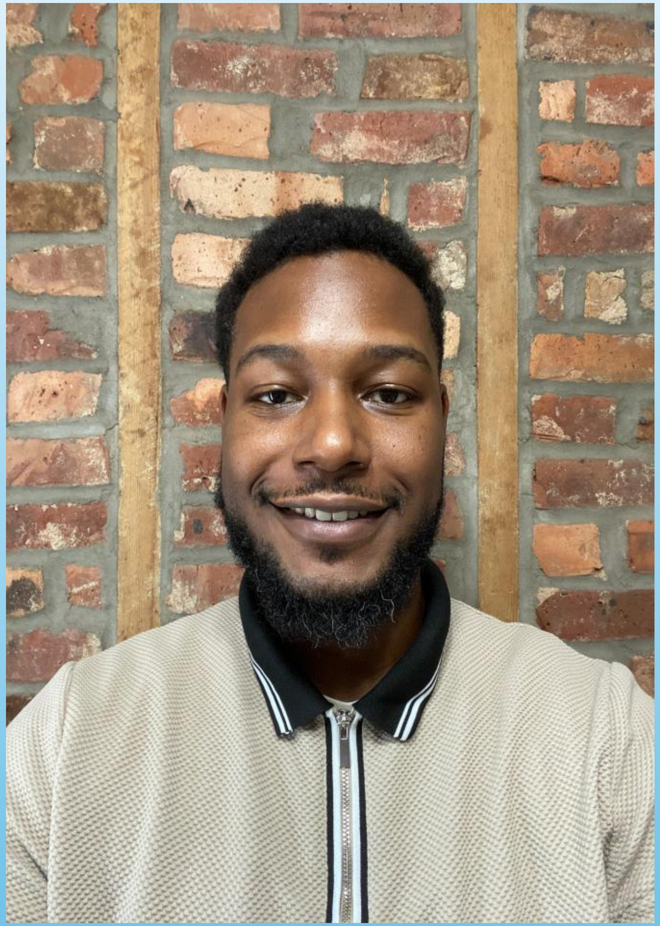
Mr. Tyler Helmrich

Role: Scrum Master



Mr. Jay Patel

Role: Back-End Developer



Mr. Xavier Jackon

Role: Machine Learning Lead

Mr. Zenil Moradia

Role: Quality assurance

Improvement from Professor

- We have improved on product backlog and user stories

Problem Statement

Credit card fraud remains the leading form of identity theft in the United States, with 2024 seeing over 449,000 complaints to the Federal Trade Commission and within the first quarter of 2025 adding another 151,000 cases. In the United States, 63% of cardholders have been targeted at least once, and more than half of these victims report experiencing repeated attacks.

This persistent threat not only inflicts hundreds of millions of dollars in annual losses but also erodes consumer confidence in digital payments. Existing detection systems struggle to keep up with the evolving tactics of scammers and the extreme imbalance between legitimate and fraudulent transactions. There is an urgent need for a robust, real-time solution that can accurately flag while minimizing false alarms.

Problem Description

Project Name:	"Classify" - Credit Card Fraud Detection Mobile App utilizing a Classification Algorithm Machine Learning Model
Team:	IT Architects
Project Description:	This project aims to deliver an advanced machine-learning classifier for credit-card fraud detection, designed for issuers and cardholders frustrated by today's rule-based systems and high false-alarm rates. Leveraging SMOTE to rebalance data and XGBoot for precision, we'll identify fraudulent transactions before funds leave the account, reduce false positives on legitimate purchases, and trigger real-time alerts that let cardholders act immediately. Our mission is to replace rigid, hand-tuned rules with a data-driven model that adapts to evolving fraud patterns by minimizing losses, preserving reputations, and restoring trust in digital payments through a mobile iOS application that is easily accessible. Current deterrents include rule-based fraud detection techniques, whereas our goal is to implement an AI-powered fraud-detection service that catches up to 80% of fraud attempts in real time while cutting false positives by 30%, so issuers lose less money and cardholders enjoy seamless, secure spending.
Benefit Outcomes:	Credit Card companies can expect a reduction in false alarms, monetary losses, and damage to their reputation. Credit Card holders can expect a reduction in unauthorized charges, fewer declined legitimate transactions, and greater peace of mind when using their cards as well as an increase in user satisfaction and engagement.
Github Link:	https://github.com/htmw/F2025-Team1/wiki

Team Working Agreement

- **Communication & Collaboration** - Our team will primarily communicate through **WhatsApp**, **Teams**, **Discord** and **GitHub** updates.
- Whenever code is pushed to GitHub, members will notify the team to prevent conflicts or overwrites.
- If anyone needs to adjust their task timeline or change their approach, they must inform the team beforehand.
- The last sprint communication will be focused on the fixation of bugs, integration, and readiness to deployment.

- **Meeting Routine & Workflow** - The team will meet twice weekly, typically on Saturday and Wednesday, unless additional sessions are required.
- Anyone unable to attend a meeting must notify the team in advance and go through the meeting summary afterward.
- Meetings will focus on -
- Feature completion
- Sprint review and final demo preparation

Team Working Agreement Cont.

- **Team Working Principles** - All tasks will include a clear deadline and will be reviewed once completed.
- All contributions should be pushed to GitHub regularly with meaningful commit messages.
- Complex tasks will be broken into smaller actionable items to balance workload.

- **Checklist for Marking a Task as Complete** - The output should satisfy all acceptance criteria, such as correct preprocessing, expected model accuracy, or UI functionality.
- All associated documentation, dataset changes should be updated.

Persona 1: Sarah Miller

- Age: 28
- Occupation: Graphic Designer

- Sarah is an avid online shopper who loves finding unique items from small businesses. She uses her credit card for most of her purchases, appreciating the convenience and rewards. She's tech-savvy and expects seamless digital experiences. Recently, her card details were stolen from a third-party website, leading to several unauthorized small purchases. The process of getting a new card and updating her payment information on various subscriptions was a significant hassle and made her wary of future online transactions. She's looking for reassurance and security in her digital financial interactions.
- Goals:
  - Wants to shop online without constant worry about card fraud.
  - Needs quick and clear notifications for any suspicious activity on her accounts.
  - Hopes for an easy way to verify transactions or freeze her card directly from her phone if needed.
- Frustrations:
  - The time and effort involved in cancelling cards and updating payment details after fraud.
  - Feeling vulnerable and exposed when her financial information is compromised.
  - Receiving generic, unhelpful alerts that don't provide clear next steps.
- Needs from the Solution:
  - Real-time alerts that are easy to understand and act upon.
  - An intuitive mobile interface that allows her to quickly approve or deny a transaction and temporarily lock her card.
  - Minimal false positives on her legitimate purchases so she doesn't lose trust in the system.

Persona 2: Alex Chen

- Age: 32
- Occupation: Founder of "PetPals Boutique" | The E-commerce Startup Founder

- Age: 32
- Occupation: Founder of "PetPals Boutique" | The E-commerce Startup Founder
- Alex recently launched "PetPals Boutique," an online store selling custom pet accessories. Business is booming, but he's starting to see a disturbing trend of chargebacks due to fraudulent orders. These chargebacks not only cost him money but also damage his store's reputation with payment processors. He's passionate about his business but has limited resources and can't afford a dedicated fraud team. He needs automated, intelligent system that can protect his revenue and allow him to focus on growing PetPals Boutique, not fighting fraud.
- Goals:
    - To minimize chargebacks and financial losses from fraudulent transactions.
    - To protect his business's reputation with payment providers.
    - To have a simple, affordable, and effective fraud detection solution that integrates easily with his e-commerce platform.
- Frustrations:
    - The financial impact of fraudulent orders and chargeback fees.
    - The time-consuming process of manually reviewing suspicious orders.
    - Fear that his small business could be targeted by organized fraud rings, which he feels ill-equipped to handle.
- Needs from the Solution:
    - An easy-to-integrate API that works seamlessly with his e-commerce platform.
    - A dashboard that provides clear, actionable insights on potential fraud, without requiring a lot of technical expertise.
    - Automated decision-making that blocks high-risk transactions instantly and allows low-risk ones to proceed without delay.

Persona 3: Marcus Johnson

- Age: 49
- Occupation: Head of Fraud Prevention at "Secure Bank"

- Marcus leads the fraud prevention department at a mid-sized bank. His team handles thousands of credit card transactions daily, constantly battling sophisticated fraud schemes. With new tactics emerging all the time, his biggest challenge is staying ahead of the criminals while ensuring legitimate customer transactions are processed smoothly. He's looking for a powerful, scalable solution that can integrate with their existing systems, reduce false positives, and provide actionable intelligence to his team. He needs to improve detection rates, reduce operational costs associated with fraud investigation, and enhance customer trust.

- Goals:
  - To significantly reduce the bank's annual losses from credit card fraud.
  - To decrease the rate of false positives, improving customer experience and reducing operational overhead.
  - To implement a system that provides real-time, adaptive fraud detection capable of learning new patterns.

- Frustrations:
  - The constant arms race with fraudsters and the difficulty of keeping up with evolving tactics.
  - The pressure to maintain a high level of security without inconveniencing legitimate customers.
  - The sheer volume of data and the challenge of sifting through it to find genuine fraud.

- Needs from the Solution:
  - A highly scalable system that can process millions of transactions in real time.
  - Advanced machine learning capabilities that can adapt to new fraud patterns and minimize false positives.

MVP

- **Real-Time Fraud Predictions –** System analyses incoming transactions instantly.
- **Interactive Dashboard –** Users can visualize fraud alerts, risk scores, and patterns through graphs & metrics.
- **Simple, Accessible UI –** Easy for non-technical users to understand fraud insights at a glance.
- **API-Driven Architecture –** Front-end fetches predictions seamlessly through backend REST APIs.
- **Notification/Alert System –** Email/SMS alerts for high-risk transactions.

Technology and Tools

Data Storage, Collection, and Processing

Kaggle – Online repository of public datasets. We used this to acquire the training data set for our model.

Pandas / NumPy – Core tabular and numerical routines.

Scikit-learn – Standardized pipeline, preprocessing, and model-selection tools.

Supabase – an open-source backend-as-a-service (BaaS) platform that provided us with a full suite of tools—database, authentication, storage, real-time subscriptions, and serverless functions.

Model Development and Experimentation

XGBoost – Gradient boosting implementations with GPU support for speed and advanced class-imbalance handling.

Python – widely used language for AI model development because of its simplicity, readability, and powerful ecosystem of libraries and frameworks.

FastAPI – a modern, high-performance Python web framework designed for building APIs quickly and efficiently.

Visualization and End-User Interface

PenPot – a free, open-source, web-based design and prototyping tool built for collaboration between designers and developers.

XCODE – Apple's official IDE for building apps across all Apple platforms.

Swift – Apple's modern, powerful, and intuitive programming language designed for building Apple Platform apps on XCODE.

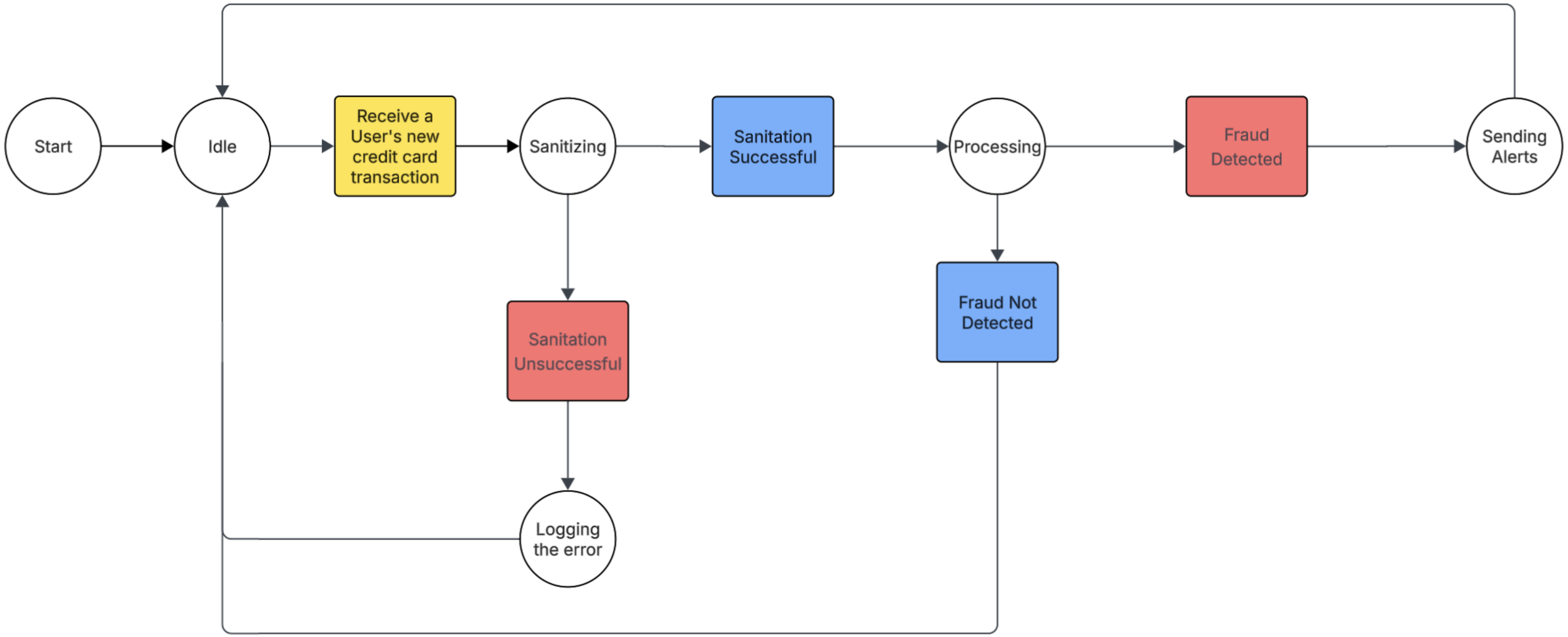
Algorithms and Models

Problem Solver	**SMOTE**	**Synthetic Minority Over-sampling Technique.** Used to mathematically generate new fraud examples on the **training data** to address the severe **class imbalance** (99.83% legitimate) and ensure the model learns fraud patterns.
Classifier	**XGBoost**	The **eXtreme Gradient Boosting** algorithm. Selected for its strength with **tabular data** and internal mechanisms to effectively maximize **Recall** (catching fraud) and maintain high **Precision** (minimizing false positives).

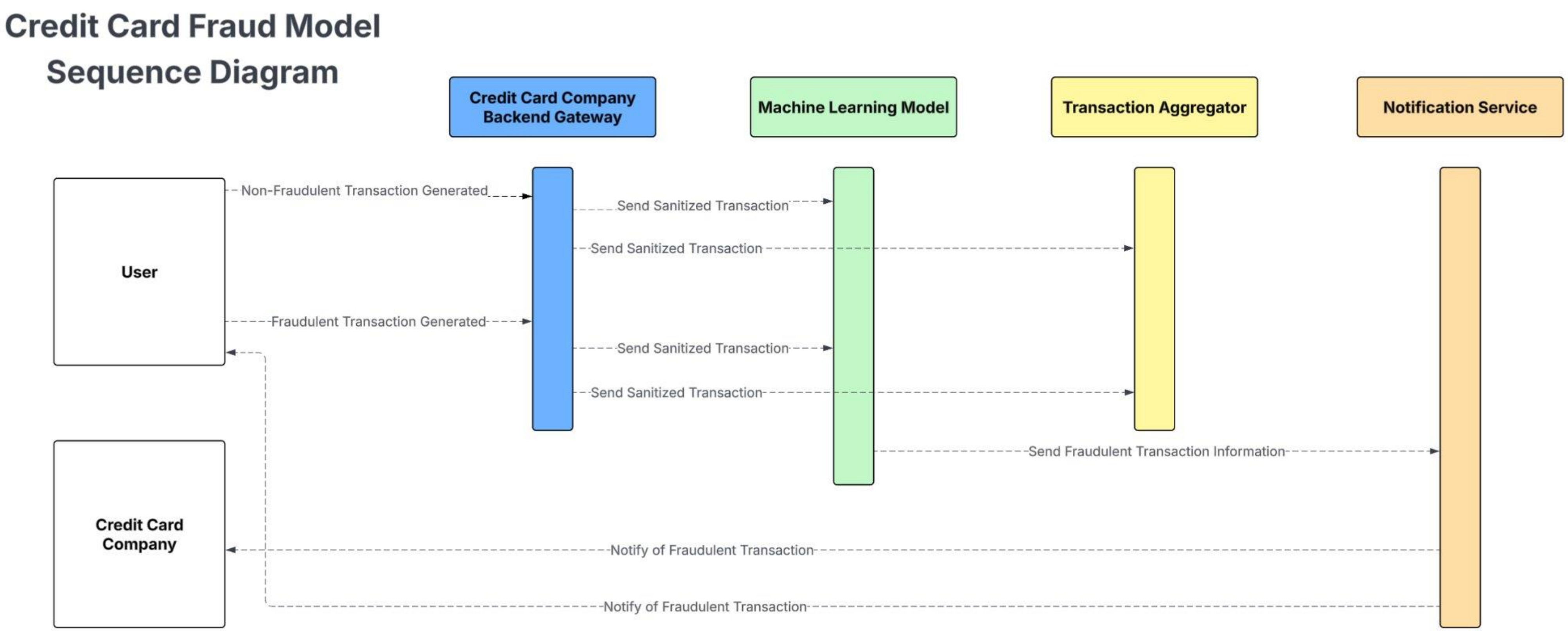
Diagrams

State Diagram

Credit Card Fraud Model State Diagram

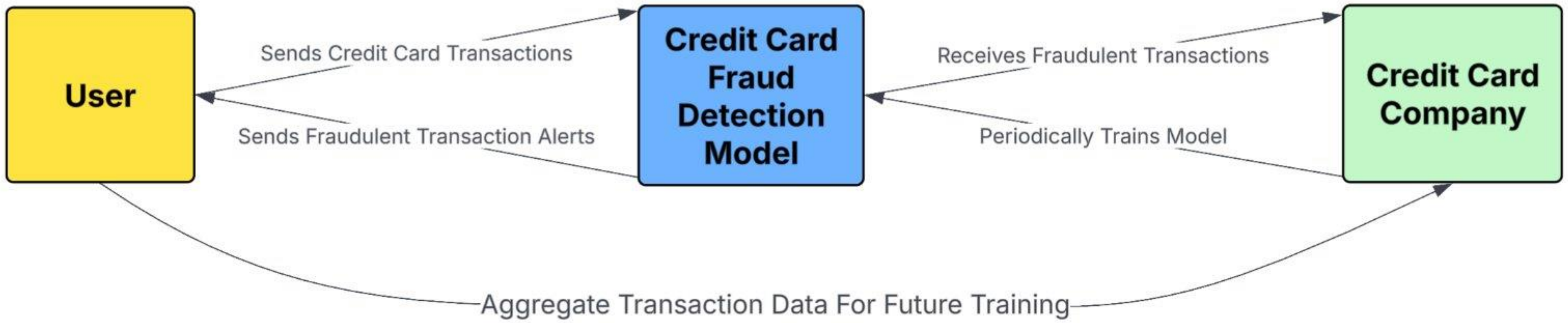


Sequence Diagram



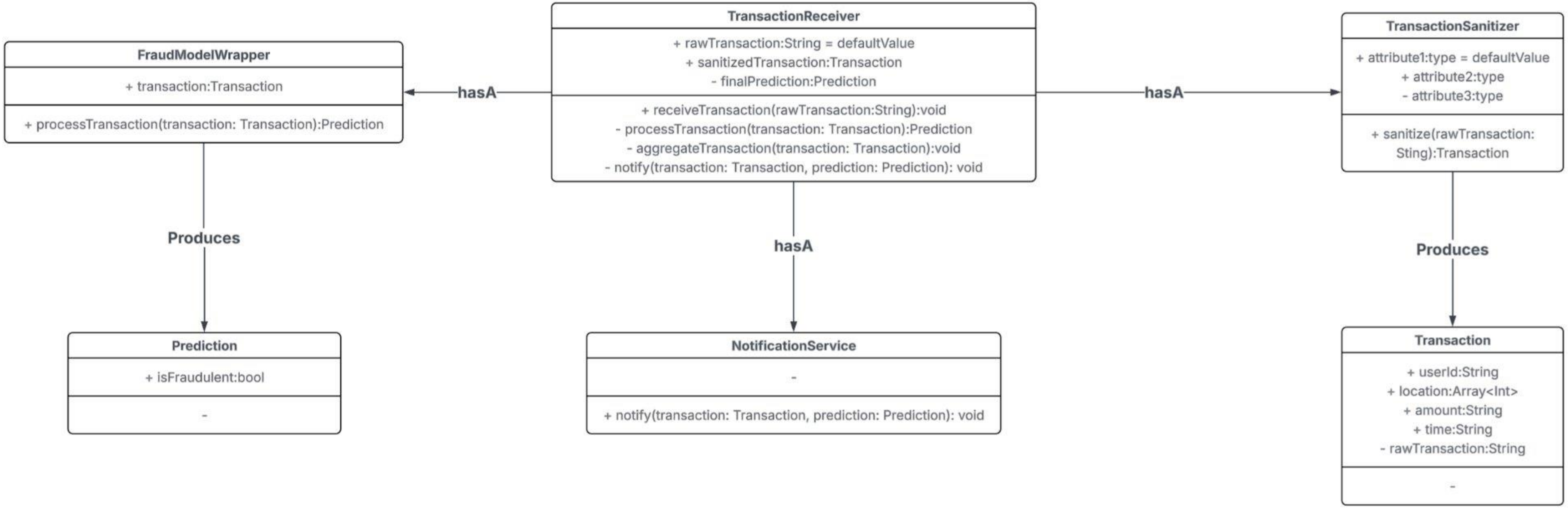
Context Diagram

Credit Card Fraud Model Context Diagram



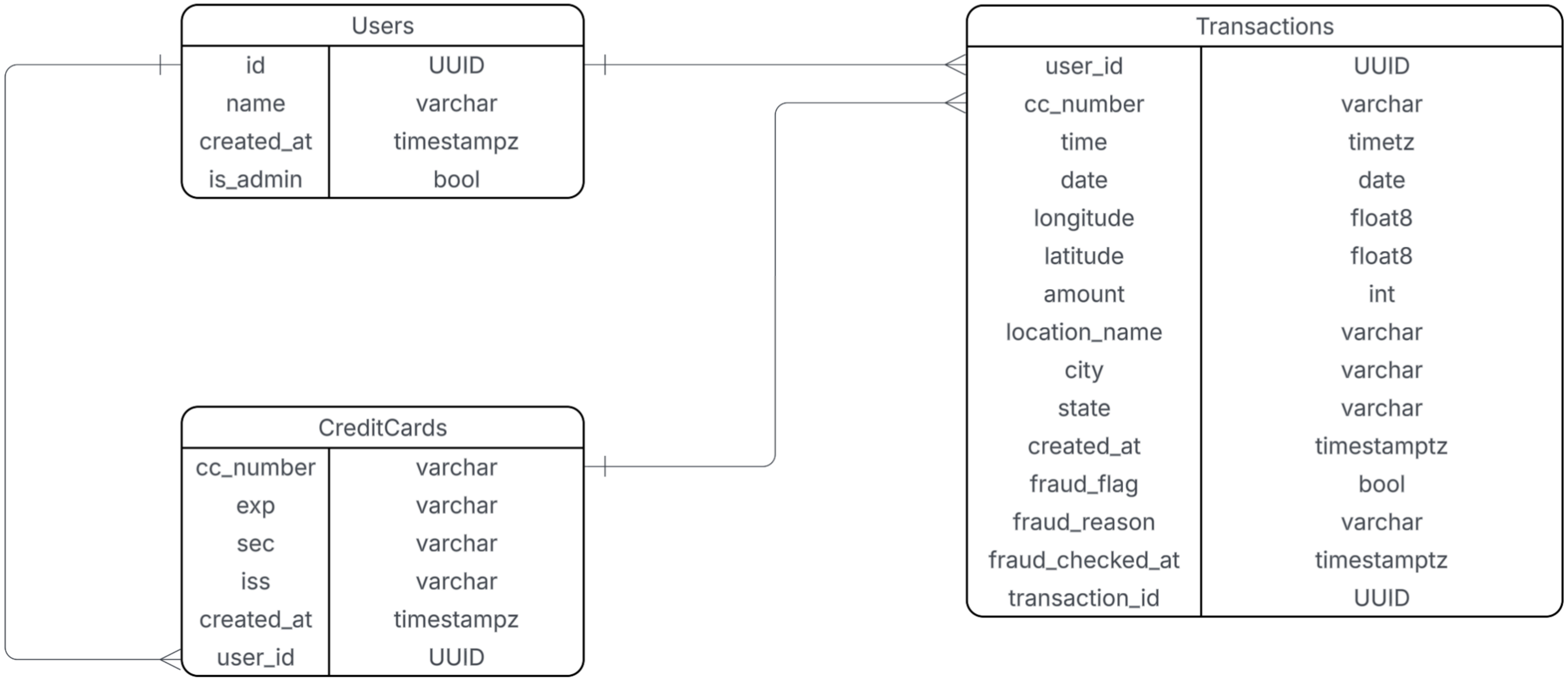
Class Diagram

Credit Card Fraud Model Class Diagram



Entity-Relationship Diagram

Credit Card Fraud
Entity-Relationship Diagram



Sprint-2 Recap

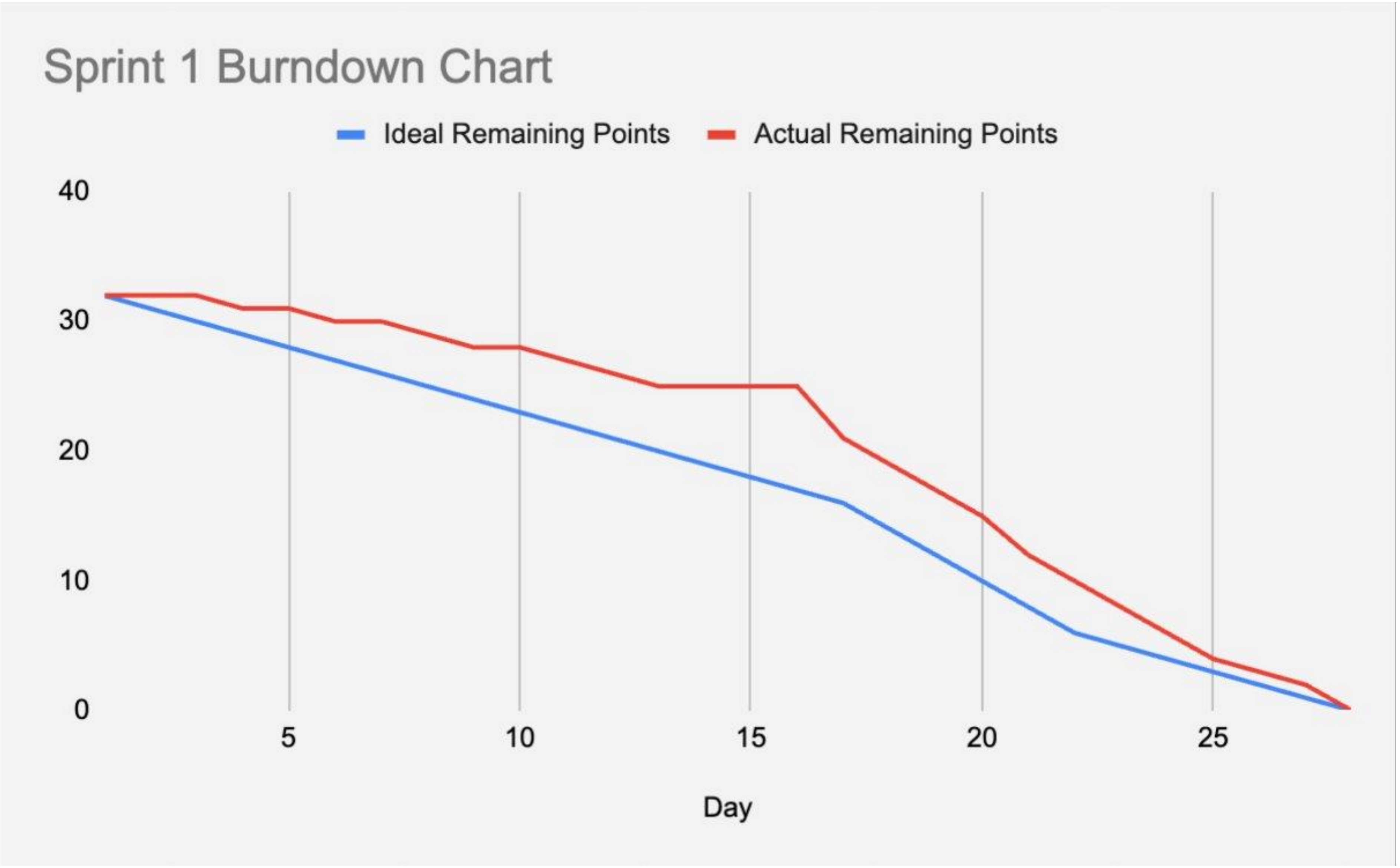
- During Sprint-2, we focused on improving the initial fraud detection model developed in Sprint-1 and expanding the overall system functionality.

- We enhanced data preprocessing and feature handling and evaluated the model’s performance using relevant metrics to ensure reliable fraud detection.

- The model was integrated more smoothly with the transaction generator, enabling real-time transaction analysis with improved input control.

- Our live demo demonstrated a complete fraud detection pipeline, from transaction generation to real-time classification of fraudulent and legitimate transactions.

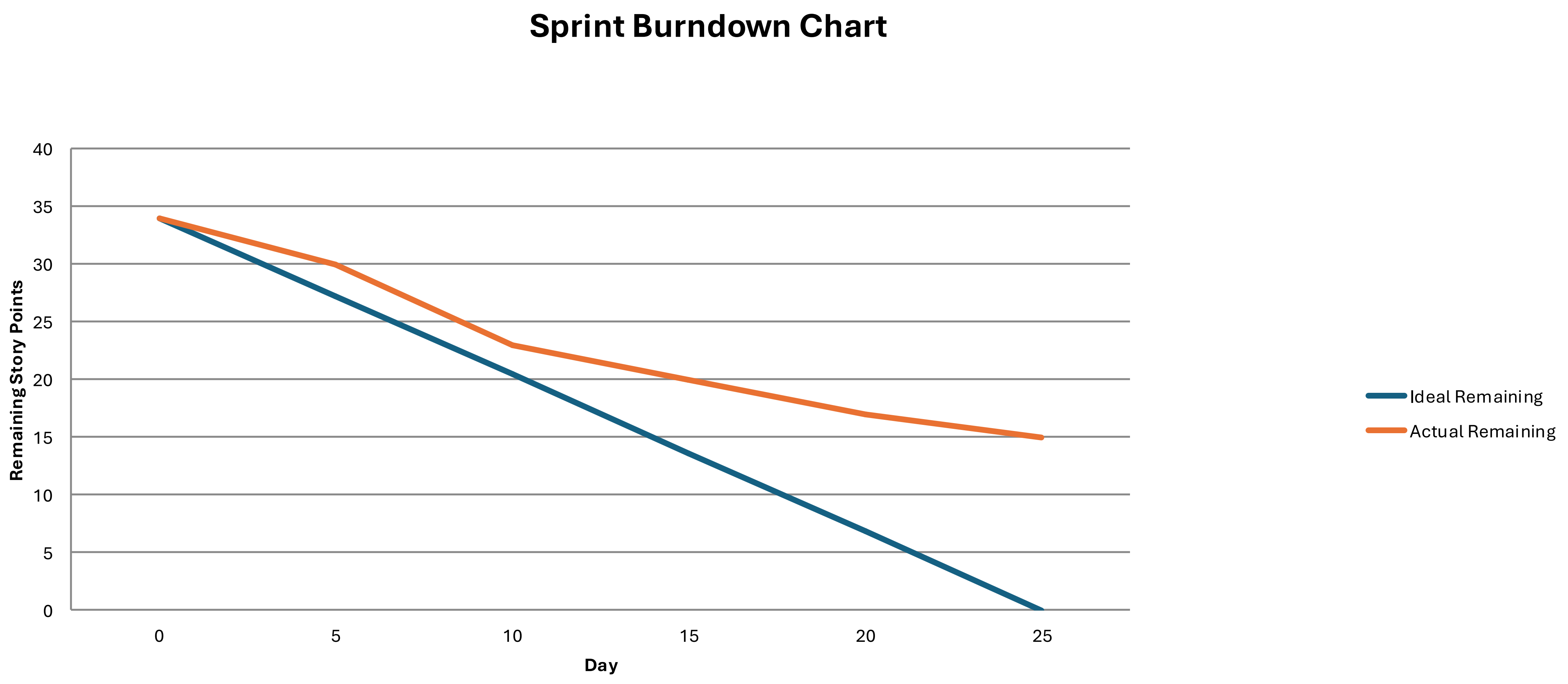
Sprint-1 Summary: Burndown Chart



Sprint-1 Summary: Stories Completed And Not Completed

S.NO	User Story	Story Points	Completed	Date Completed
1.	As a system owner I want to make a base model for prediction	8	8	-
2.	As a system owner, I want to make a demo UI model for testing and development	10	10	-
3.	As a system owner base research needs to be done to understand how to develop the machine learning model	10	10	
4.	As a system owner we need to find a good source of data in order to train our model	4	4	

Sprint-2 Summary: Burndown Chart



Sprint-2 Summary: Stories Completed And Not Completed

S.NO	User Story	Story Points	Completed	Date Completed
US_02	As a user, I want to link my credit card so that the system can start monitoring transactions for fraud.	3	3	Nov 4
US_03	As the system, I want to ingest transactions so that fraud analysis can begin immediately.	5	5	Nov 12
US_15	As an admin, I want analytic reports so that I can review overall fraud trends monthly.	5	0	-
US_06	As the system, I want to run a machine learning fraud detection model so that fraudulent patterns can be identified.	8	8	Nov 16
US_07	As a fraud analyst, I want to mark transactions as fraud or safe so that the model improves over time.	3	3	Nov 12
US_10	As an admin, I want to manage system roles so that access controls are enforced.	3	0	–

Sprint-2 Summary:Stories Completed And Not Completed

S.NO	User Story	Story Points	Completed	Date Completed
US_12	As a user, I want to file a dispute for fraudulent charges so the bank can take action.	3	0	-
US_13	As the system, I want to maintain audit logs so that all actions related to fraud can be reviewed.	2	0	-
US_14	As a user, I want to enable/disable fraud notifications so I have control over alerts.	2	0	-

Sprint-3 Product Backlog

S.NO	User Story	Acceptance Criteria	Story Points
US_01	As a Registered user, I want to securely register and log in so that my financial data remains protected.	User can sign up, log in, and authentication is validated using encrypted credentials and session tokens.	5
US_02	As System user, I want to link my credit card so that the system can analyze transactions for fraud.	User can add, validate, and save encrypted card details.	5
US_03	As a developer, I want to ingest real-time transactions so that fraud detection happens instantly.	System successfully receives, validates, and stores transaction streams in the database.	8
US_04	As a fraud analyst, I want to review risky transactions so that I can quickly take action.	Analysts can view flagged transactions with risk scores and transaction details.	5
US_05	As a product user, I want to receive alerts when suspicious activity occurs so that I can respond immediately.	Alerts trigger when fraud score exceeds threshold via dashboard notification.	5

Sprint-3 Product Backlog

S.NO	User Story	Acceptance Criteria	Story Points
US_06	As the system, I want to run ML models to detect fraudulent patterns so that high-risk transactions are identified.	ML model generates fraud probability score for every transaction.	8
US_07	As an admin, I need to secure access to my dashboard so that no unauthorized person can view sensitive data.	Admin dashboard is protected with role-based access control and audit logging.	5
US_08	As a fraud analyst, I want a rule-based engine so I can define custom fraud rules.	Analysts can create and activate rules such as amount limits or location mismatch.	5
US_09	As a product user, I want to view spending analytics so I can understand my financial behavior.	Dashboard displays charts for spending patterns, merchants, and time-based trends.	3
US_10	As a fraud analyst, I want to mark transactions as fraud or safe for model learning.	Analyst can update labels, add notes, and change case status.	5

Sprint-3 Product Backlog

S.NO	User Story	Acceptance Criteria	Story Points
US_11	As a system user, I want to file a chargeback for confirmed fraudulent transactions.	User can generate a chargeback request with required transaction information.	5
US_12	As an admin, I want to manage and verify financial institutions integrating with our system.	Admin can approve or reject banks after verifying API credentials.	5
US_13	As the system Developer, I want to log all fraud-related events so that investigators can review them later.	Every fraud-related action is saved with timestamp, reviewer notes, and device metadata.	3
US_14	As a product user, I want OTP verification for high-risk login attempts so that my account stays protected.	OTP is sent on suspicious logins and must be verified to continue.	5

Sprint-3 User backlog

S.NO	User Story	Acceptance Criteria	Feature	Story Points
US_02	As a user, I want to link my credit card so that the system can start monitoring transactions for fraud.	User can add a card, Luhn validation passes, and card is stored securely.	Credit Card Linking	3
US_03	As the system, I want to ingest transactions so that fraud analysis can begin immediately.	System receives, validates, and stores transaction data automatically.	Transaction Ingestion	4
US_04	As a user, I want to receive alerts when suspicious activity occurs so that I can take action quickly.	Alert triggers when fraud score exceeds threshold.	Fraud Alerts	3
US_05	As a user, I want the application to be secure so that my data remains safe	User data is stored securely	Audit logs / security monitoring	13

Sprint-3 User backlog

S.NOs	User Story	Acceptance Criteria	Feature	Story Points
US_06	As a user, I want a visually appealing and user-friendly interface so that I can easily navigate and enjoy using the application.	The interface uses a clean and consistent design.	User friendly interface	13
US_07	As a fraud analyst, I want to mark transactions as fraud or safe so that the model improves over time.	Analyst can label transactions and add case notes.	Case Management	3
US_08	As a user, I want to view my recent transactions so I can verify if anything looks suspicious.	User can view transaction list with amount, date, merchant, status.	Transaction History	8

Sprint-3 User backlog

S.NO	User Story	Acceptance Criteria	Feature	Story Points
US_09	As a user, I want to file a dispute for fraudulent charges so the bank can take action.	User fills dispute form linked to a specific transaction.	Fraud Dispute Filing	3
US_10	As a user, I want the system to monitor all my cards so that I can stay informed and protected.	The system monitors activity on all user-linked cards.	Real-time card activity tracking	14
US_11	As a user, I want to enable/disable fraud notifications so I have control over alerts.	User can toggle notification preferences in settings.	Notification Settings	3

Sprint-3Test Cases

US_ID	TC_ID	Test Case	Steps	Execution Status	Result
US_05	TC_31	Trigger Fraud Alert for Suspicious Transaction	1. Login as Customer 2. Perform high-value transaction 3. Check system fraud alert 4. Validate alert is logged	Fraud alert should trigger	Pass
US_06	TC_33	Approve/Decline Suspicious Transaction	1. Login as Analyst 2. Open fraud alert 3. Approve/Decline 4. Add notes 5. Submit	Decision recorded	Pass
US_12	TC_39	Export Monthly Fraud Report	1. Login as Admin 2. Go to Reports 3. Generate Monthly Report 4. Download PDF/Excel	Report exported	Pass
US_09	TC_36	Customer Receives Fraud Notification	1. Trigger fraud 2. System blocks card 3. Notification sent SMS/Email 4. Verify message	Notification received	Pass

Sprint-3 Test Cases

US_ID	TC_ID	Test Case	Steps	Execution Status	Result
US_09	TC_36	Customer Receives Fraud Notification	1. Trigger fraud 2. System blocks card 3. Notification sent SMS/Email 4. Verify message	Notification received	Pass
US_10	TC_37	False Positive Check – Legit Transaction	1. Perform normal transaction 2. Model evaluates 3. No fraud alert 4. Transaction succeeds	Transaction allowed	Pass
US_11	TC_38	Fraud Analyst Cannot View Restricted Cases	1. Login as Analyst A 2. Try accessing Analyst B’s case 3. Access should be blocked	Unauthorized access blocked	Fail

Stories Completed

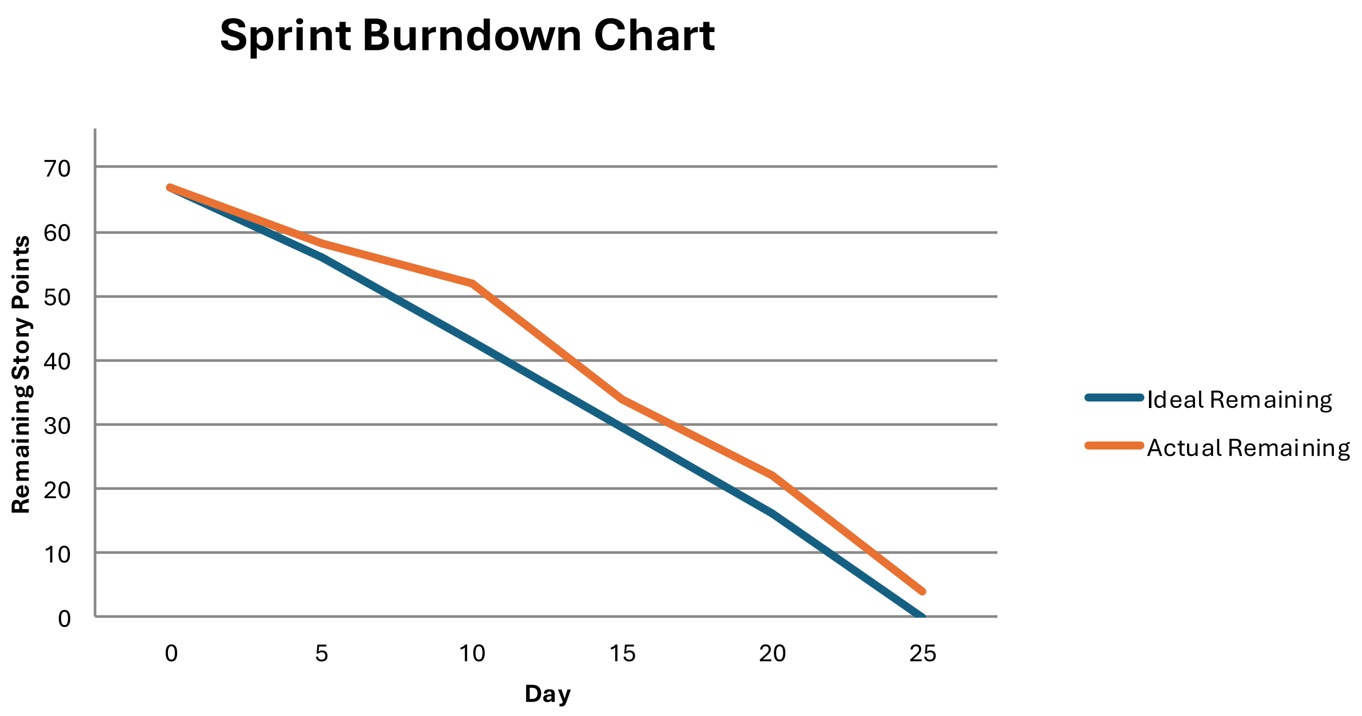
S.NO	User Story	Story Points	Completed	Date Completed
US_02	As a user, I want the application to be secure so that my data remains safe	13	13	Dec 4
US_06	As a user, I want a visually appealing and user-friendly interface so that I can easily navigate and enjoy using the application.	13	13	Dec 5
US_15	As an admin, I want analytic reports so that I can review overall fraud trends monthly.	5	5	Dec 5
US_10	As a user, I want the system to monitor all my cards so that I can stay informed and protected.	14	14	Dec 10
US_11	As a user, I want to enable/disable fraud notifications so I have control over alerts.	3	3	Dec 7
US_10	As an admin, I want to manage system roles so that access controls are enforced.	3	3	Dec8

Stories Completed

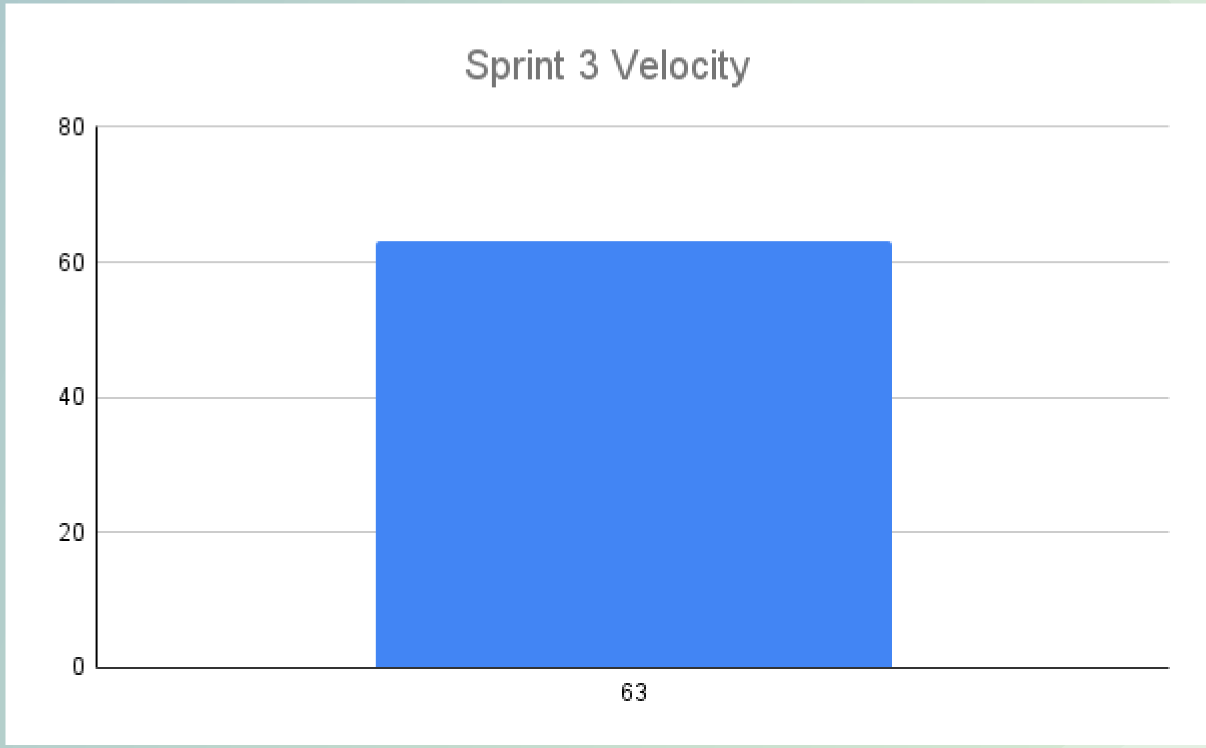
S.NO	User Story	Story Points	Completed	Date Completed
US_04	As a user, I want to receive alerts when suspicious activity occurs so that I can take action quickly.	3	3	Nov 30
US_09	As a user, I want to file a dispute for fraudulent charges so the bank can take action.	4	0	-

Metrics

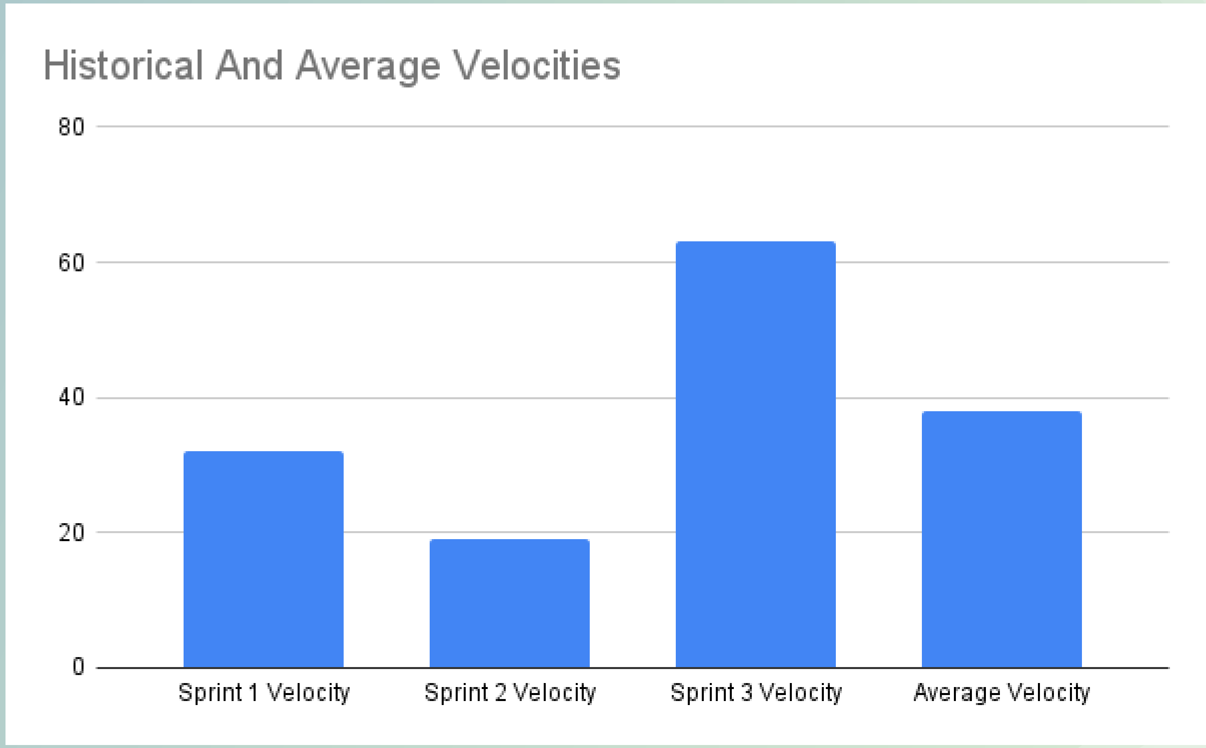
Sprint 3 Burndown Chart



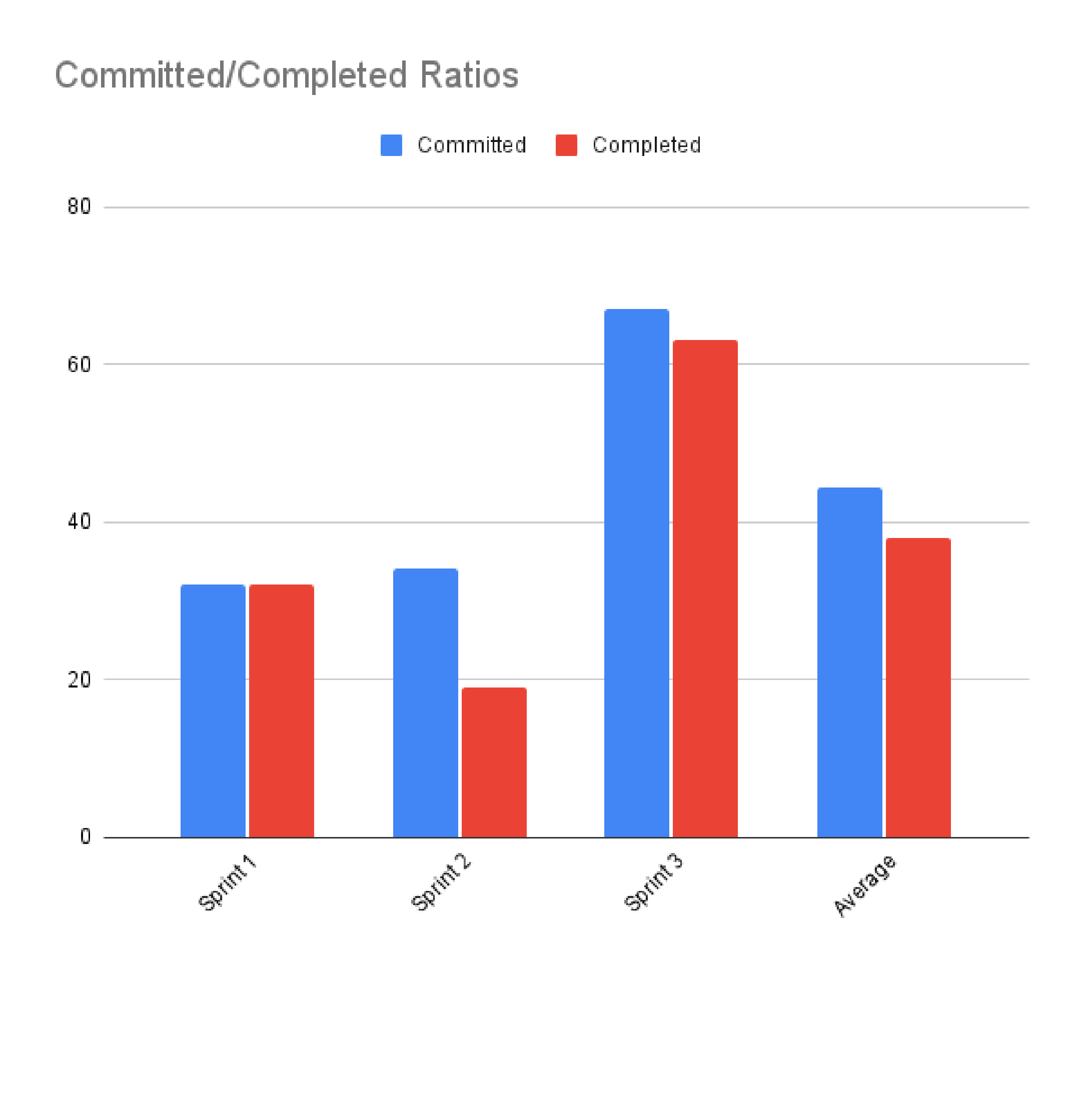
Sprint 3 Velocity Chart



Historical & Average Velocities



Committed & Completed Ratios



Sprint 3 Retrospective

- Video can be viewed here: https://youtu.be/JHORY5hOy9o

Sprint 3 Retrospective

What We Did Well	Votes
Collaborated well over discord during shared work sessions	4
Programming and debugging in pairs	3
We have figured out all the confusion we had in our sprint-2 regarding our iOS application	3
High quality demo preparation	1
Push Notifications are a good feature	1
this time we accomplished MVP that we can in this specific time period.	1
Effective use of tools	1
did given out good amount of efforts this time	1
We are able to finish up our fraudulent report page	0
Documentation creation	0
Meeting were really went well	0
good understanding we develop related this field	0

What We Can Do Differently	Votes
If given extra time to work on the project, take advantage of that time so we aren't rushing to complete things last minute	5
Our synchronous collaboration is good but work needs to be specifically dolled out	5
Notify team members on what tasks you are working on so we don't overlap on tasks	5
We need to get a better grasp on all of the project deliverables such as the user and deployment manuals so we don't leave them to the end	3
Document decisions earlier	2
improve effort estimation	1
we can decide concrete plan for our app like what things we want to do as in begining.	1
we needs to have better understanding about resources/tools we are using	0

Action Items	Votes
Have a better grasp on the deliverables based on the user stories.	0
During Sprint Planning, we need to assign user stories to people.	0
Work on documentation earlier.	0
Have more time discipline.	0

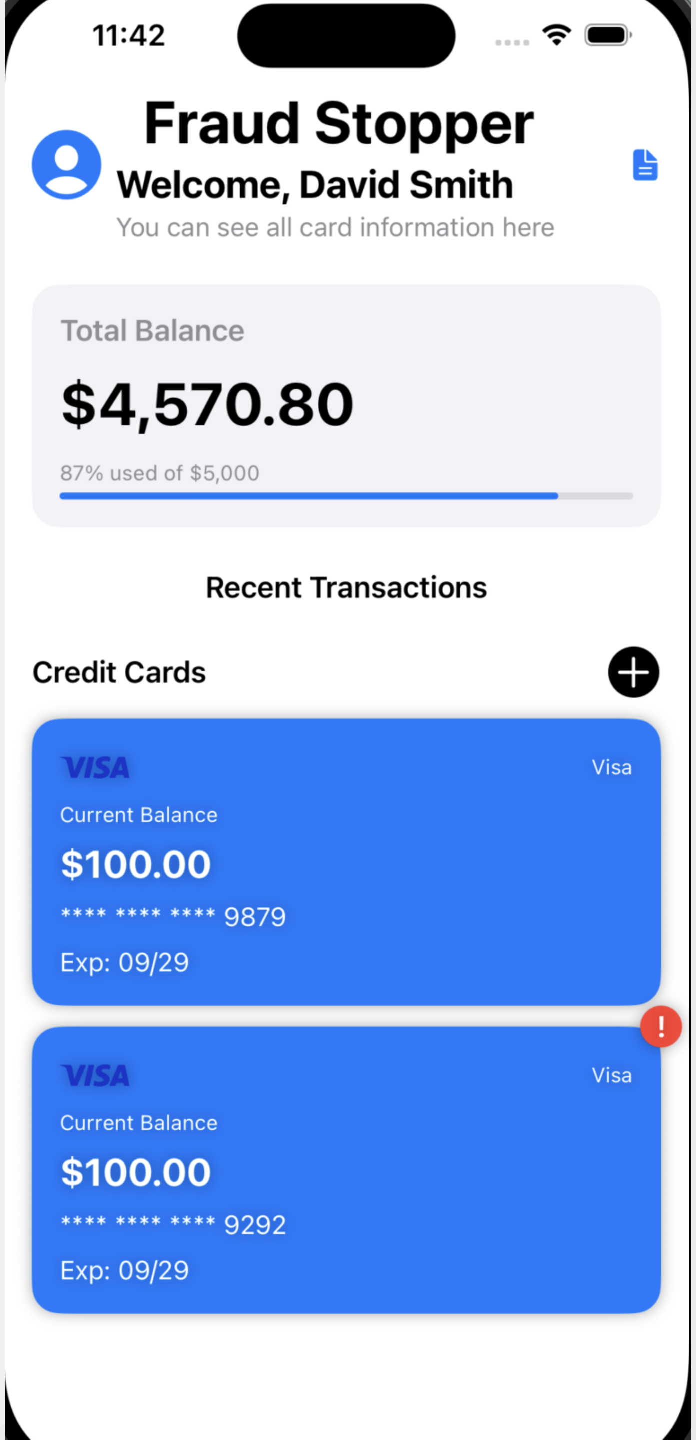
Sprint-3 Planning Stories Carried Over

S.NO	User Story	Acceptance Criteria	Feature	Story Points
US_10	As an admin, I want to manage system roles so that access controls are enforced.	Admin can assign roles (user, analyst, admin) with RBAC.	Role-Based Access Control	3
US_12	As a user, I want to file a dispute for fraudulent charges so the bank can take action.	User fills dispute form linked to a specific transaction.	Fraud Dispute Filing	3
US_13	As the system, I want to maintain audit logs so that all actions related to fraud can be reviewed.	Every important action logged with timestamp, user, device info.	Audit Log System	2
US_14	As a user, I want to enable/disable fraud notifications so I have control over alerts.	User can toggle notification preferences in settings.	Notification Settings	2
US_15	As an admin, I want analytic reports so that I can review overall fraud trends monthly.	Monthly reports generated automatically with charts.	Fraud Reporting Engine	5

Application Screenshots

- The Homepage shows a list of user credit cards, cards that have experienced fraudulent transactions will have a red exclamation badge to indicate the fraud to the user

- The User can scroll their list of credit cards and can click into a credit card to view the transactions for that card (shown on the next page)

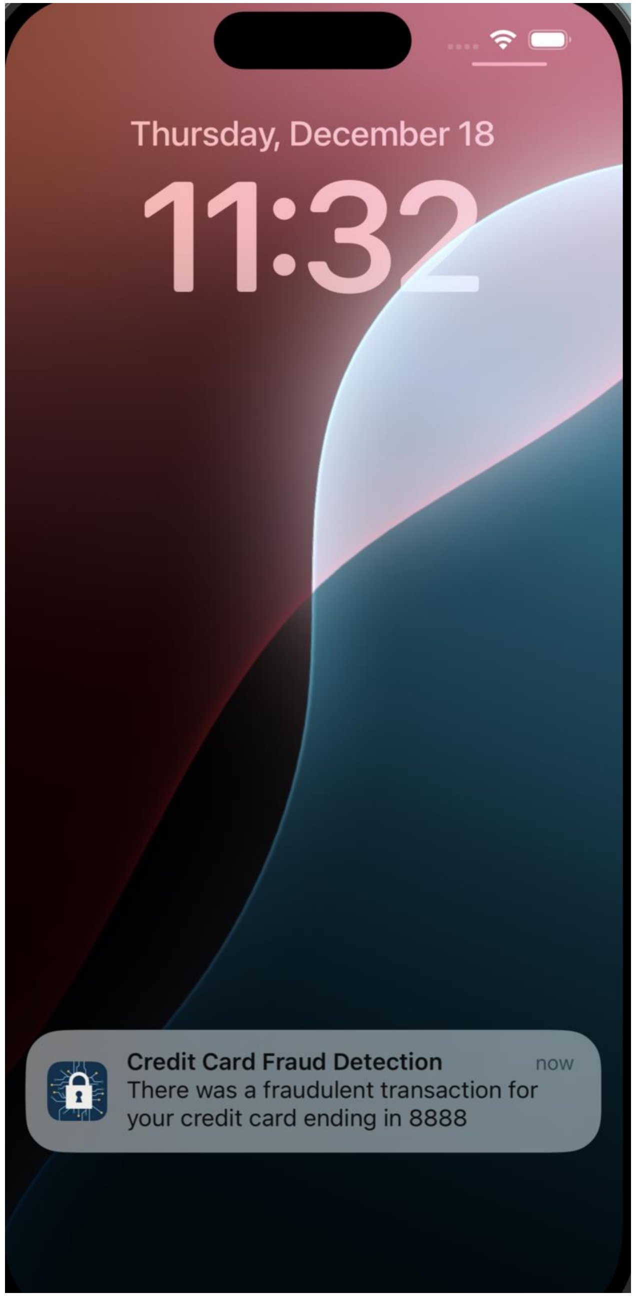


Application Screenshots

Real-Time Push Notifications: The application utilizes local notifications to instantly alert the user when the machine learning model flags a transaction as fraudulent.

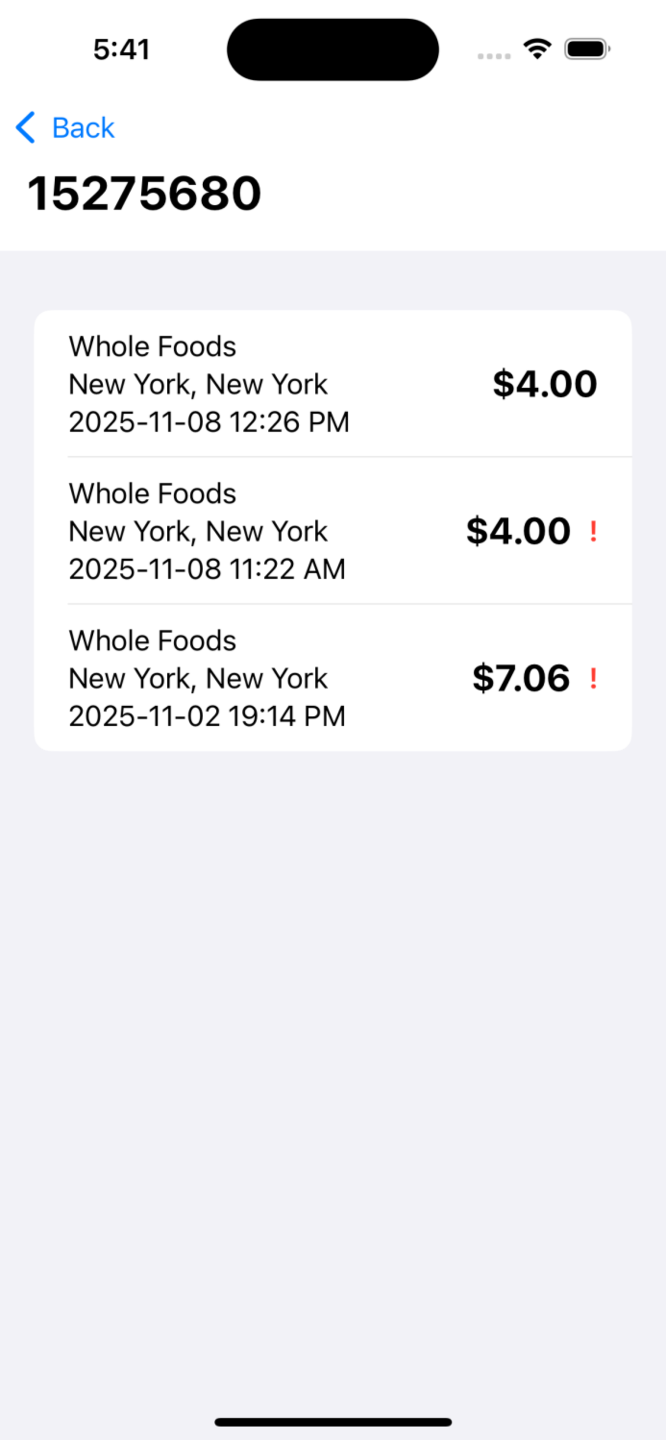
Immediate User Awareness: Ensures users are notified even when the application is closed or the phone is locked.

Contextual Information: The alert displays key details—such as the specific credit card ending in "8888"—allowing the user to recognize the card immediately.



Application Screenshots

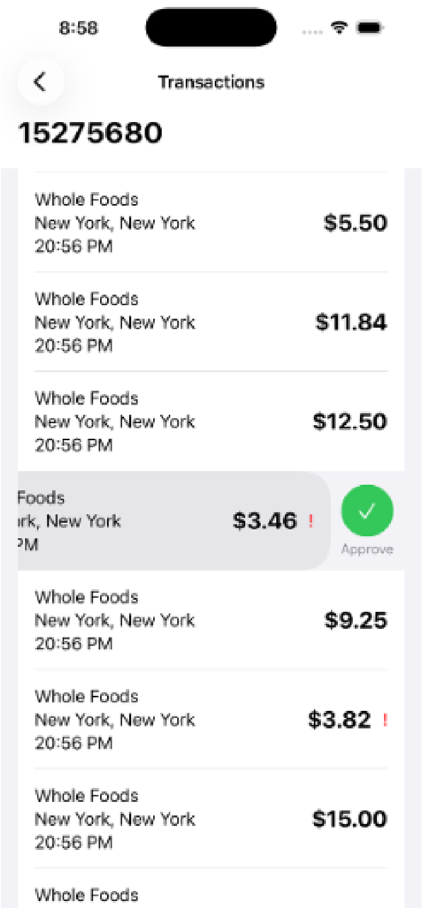
- This page shows a list of transactions for a specific credit card
- The credit card number is shown at the top of the list as a header
- A transaction row contains the location, city, state, and time of transaction along with the amount
- If a transaction is believed to be fraudulent it will have a red exclamation point to the right of its transaction amount

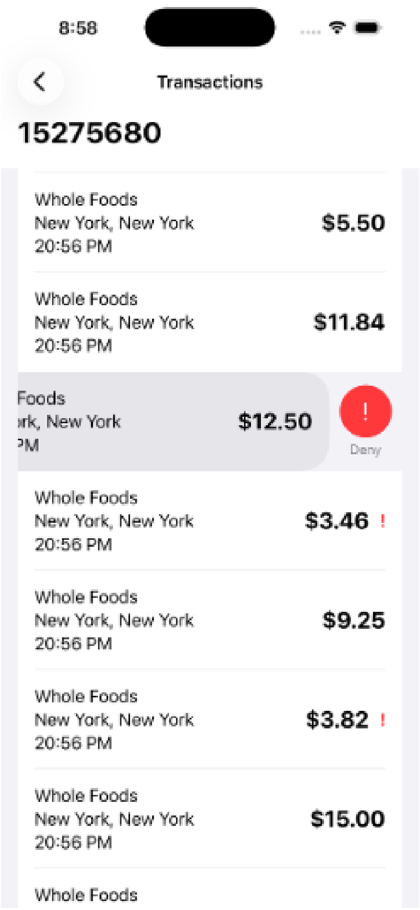


Approving and Denying Transactions

Swipe left on denied transaction → Tap green checkmark → Mark as Approved.

Swipe left → Tap red deny icon → Mark as Denied.



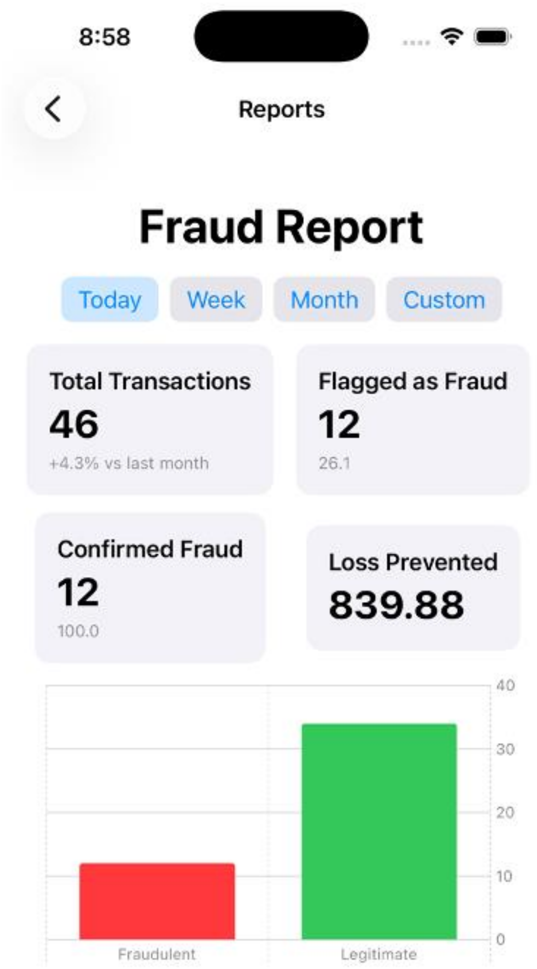


Fraud Report

The Reports Section provides analytics across different periods (Today, Week, Month and Custom).

Metrics include:
- Total Transactions: All processed transactions.
- Flagged as Fraud: Transactions identified as suspicious.
- Confirmed Fraud: Fraudulent transactions denied by the user.
- Loss Prevented: The amount saved based on detected and denied fraud cases.

Charts show a comparison of fraudulent vs legitimate activity to help understand spending patterns and risk exposure.



REST API

- All APIs routes require an API secret in the auth header in-order to access the route
- User ID information is also included in the http header rather than the URL

GET /functions/v1/app-middleware/user : Retrieves all user information such as name, email, and list of credit cards

POST /functions/v1/app-middleware/creditcard : Adds a new credit card to the user's account

```
{
    "userId": "...",
    "ccNumber": "...",
    "exp": "...",
    "sec": "...",
    "iss": "MASTERCARD | VISA | DISCOVER | etc."
}
```

GET /functions/v1/app-middleware/dailyreport?startTime=<…>&endTime=<…>

If the user is an admin they can use this route to access a daily report of fraudulent transactions

GET /functions/v1/app-middleware/transactions?ccNumber=<…>

Retrieves all transactions for a user, with the option to get transactions only for a specific credit card

REST API - Continued

POST /functions/v1/app-middleware/transactions : Create a new transaction

```
{
    "userId": "...",
    "ccNumber": "...",
    "longitude": 0.0,
    "latitude": 0.0,
    "amount": 400,
    "fraud_flag": false // false | true
}
```

PUT /functions/v1/app-middleware/transactions : Update the state of a transaction, primarily fraud status

```
{
    "fraud_flag": false // set to false | true
}
```

REST API - Continued

POST /functions/v1/app-middleware/signup: Register a new user

```
{
    "email": "tyler.helmrich+3333@gmail.com",
    "password": "1234",
    "name": "Tyler Helmrich"
}
```

GET /functions/v1/app-middleware/login?email=<email>&password=<password>: Login an existing user

REST API - Continued

POST /functions/v1/app-middleware/pns: Register a user's device for push notifications

```json
{
    "user_id": "4d16cd41-1c66-4acd-9901-707b4f0727b9",
    "token": "example_token"
}
```

POST /functions/v1/app-middleware/notify: Send a push notification to a user's device

```json
{
    "user_id": "deb92102-1adb-4be9-8460-f893ea521495",
    "text": "There was a fraudulent transaction for your credit card ending in 3838"
}
```

WIKI Page & GitHub Repository

[IT Architects Wiki Page]

[IT Architects Github Repository]

Project Demo

Thank You